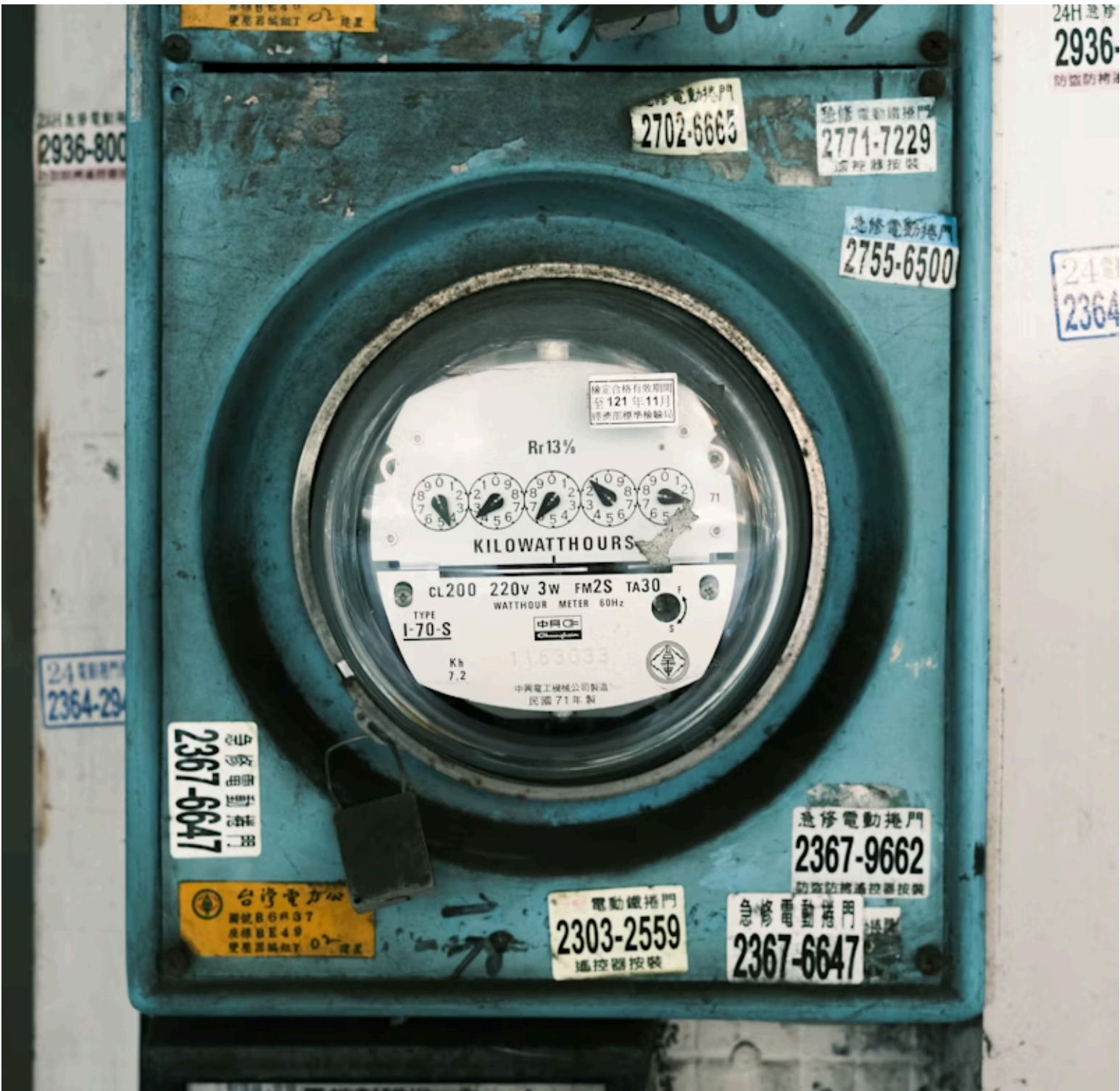




PROJECT 8

HOUSEHOLD ELECTRICITY USAGE

Multicollinearity in Regression Model

TOPIC REVIEW - COLLINEARITY

Collinearity is a statistical condition in which two independent variables (variables X) in a regression model have a very strong linear relationship with each other.

Simply put, our variables overlap excessively. This confuses the model in determining which variables are truly influential, making the resulting coefficients unreliable as a single measure of influence.

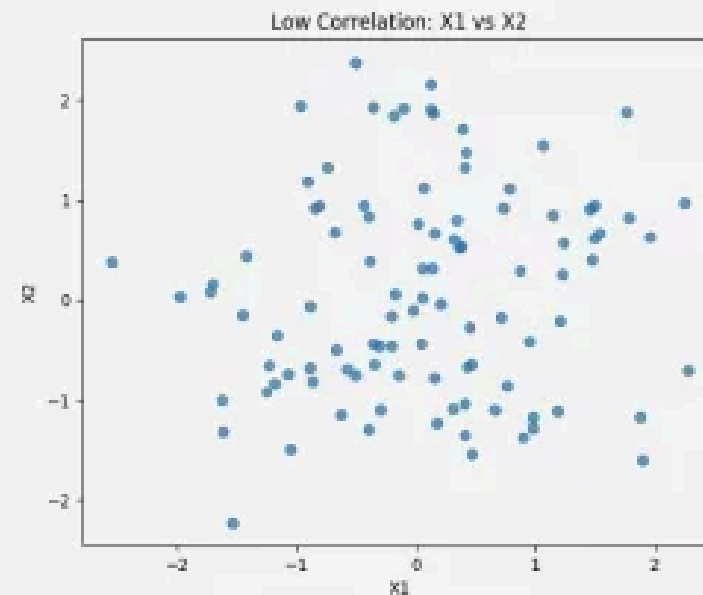
why we should avoid collinearity (multicollinearity):

- Unstable Estimates. Parameter estimation becomes unreliable and fluctuates across samples.
- Significance Loss. Variables lose statistical significance even when truly influential.
- Overlap Effect. Interpretation is blurred — model can't isolate each variable's impact.
- Overfitting Risk. Model fits noise in training data, performing poorly on new data.

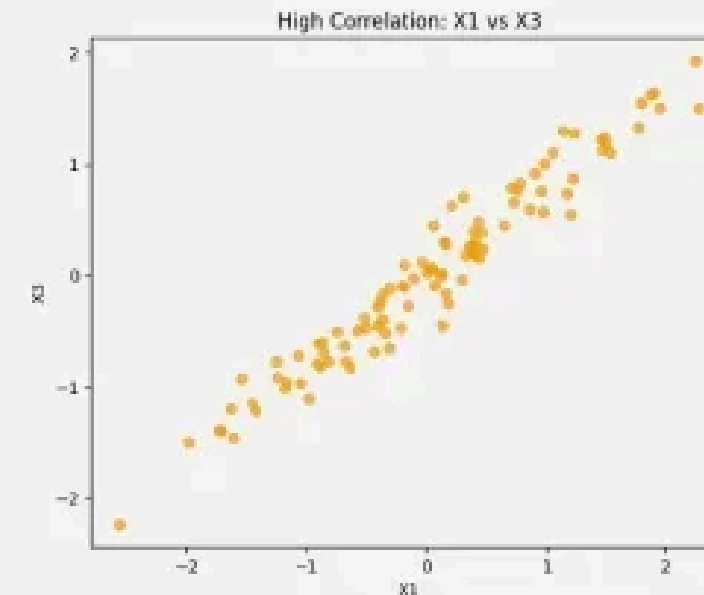
TOPIC REVIEW - COLLINEARITY

Understanding Multicollinearity

Multicollinearity causes redundancy, leading to unstable results and making interpretation difficult.



Low Correlation - No Multicollinearity



High Correlation - Multicollinearity

source : [geeksforgeeks](https://www.geeksforgeeks.org/multicollinearity/)

BACKGROUND

In this case study, I implemented an electrical regression model to analyze the use of temperature variables and other independent variables that have near collinearity with the temperature.

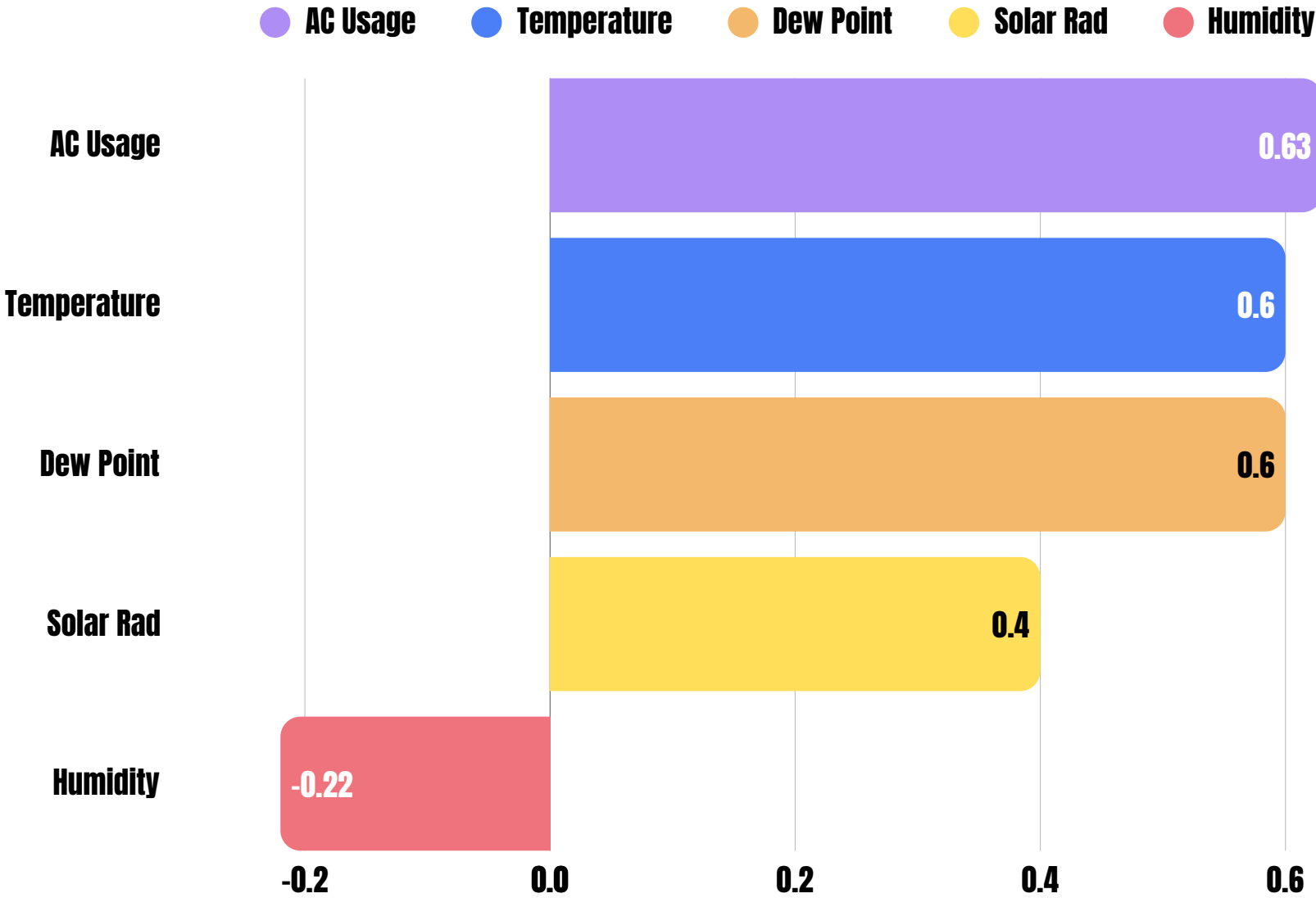
CORRELATION HEATMAP ANALYSIS

Creating a heatmap to check for multicollinearity aims to visualize the correlation between predictor variables. The goal is to identify redundant or highly correlated variable pairs before building the model.



CORRELATION HEATMAP ANALYSIS

Correlation with Electricity Usage



High Collinearity Risk

Temperature ↔ Dew Point
r = 0.995

critical

Temperature ↔ Solar Rad
r = 0.691

middle

Temperature ↔ AC Usage
r = 0.029

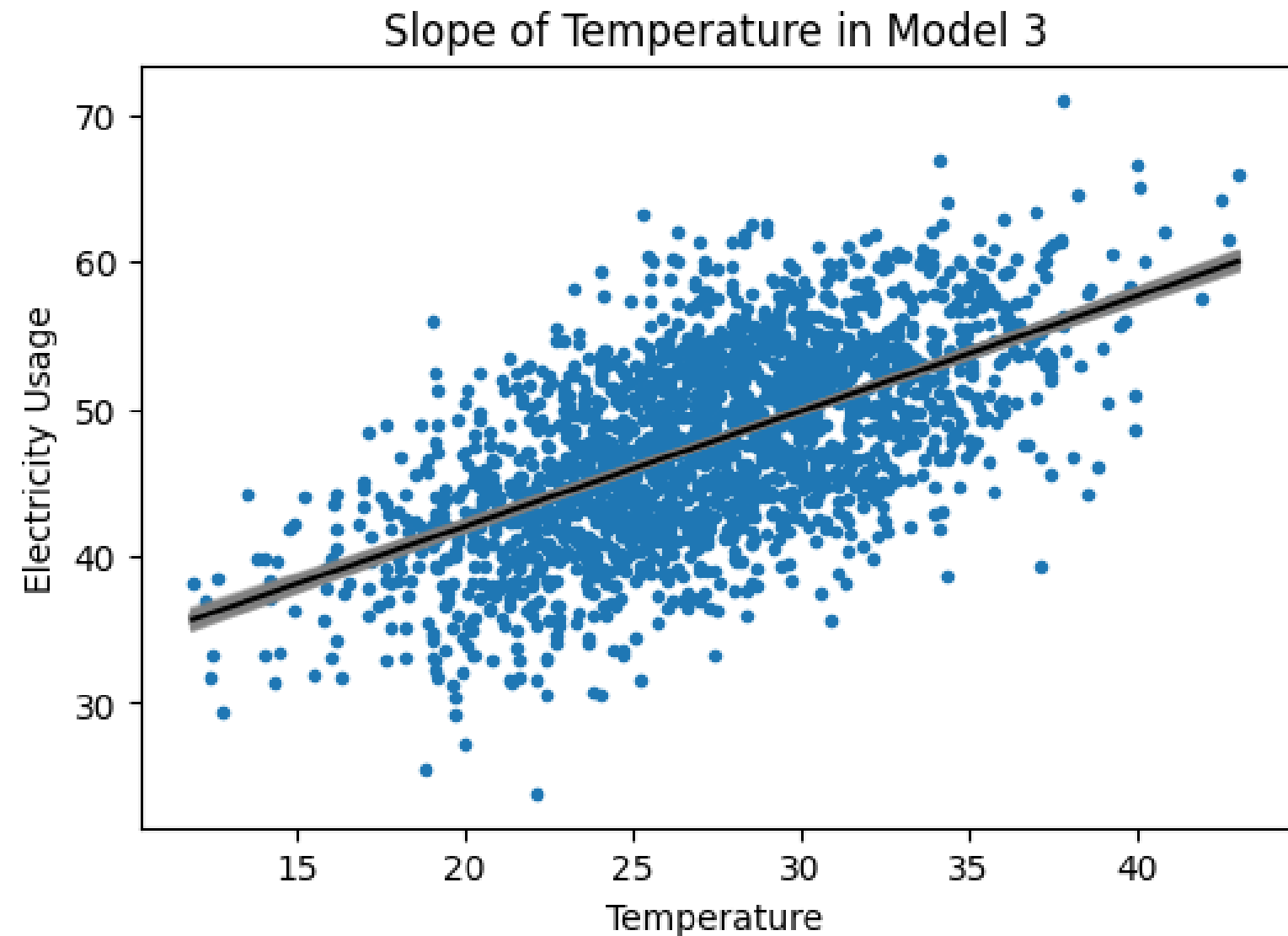
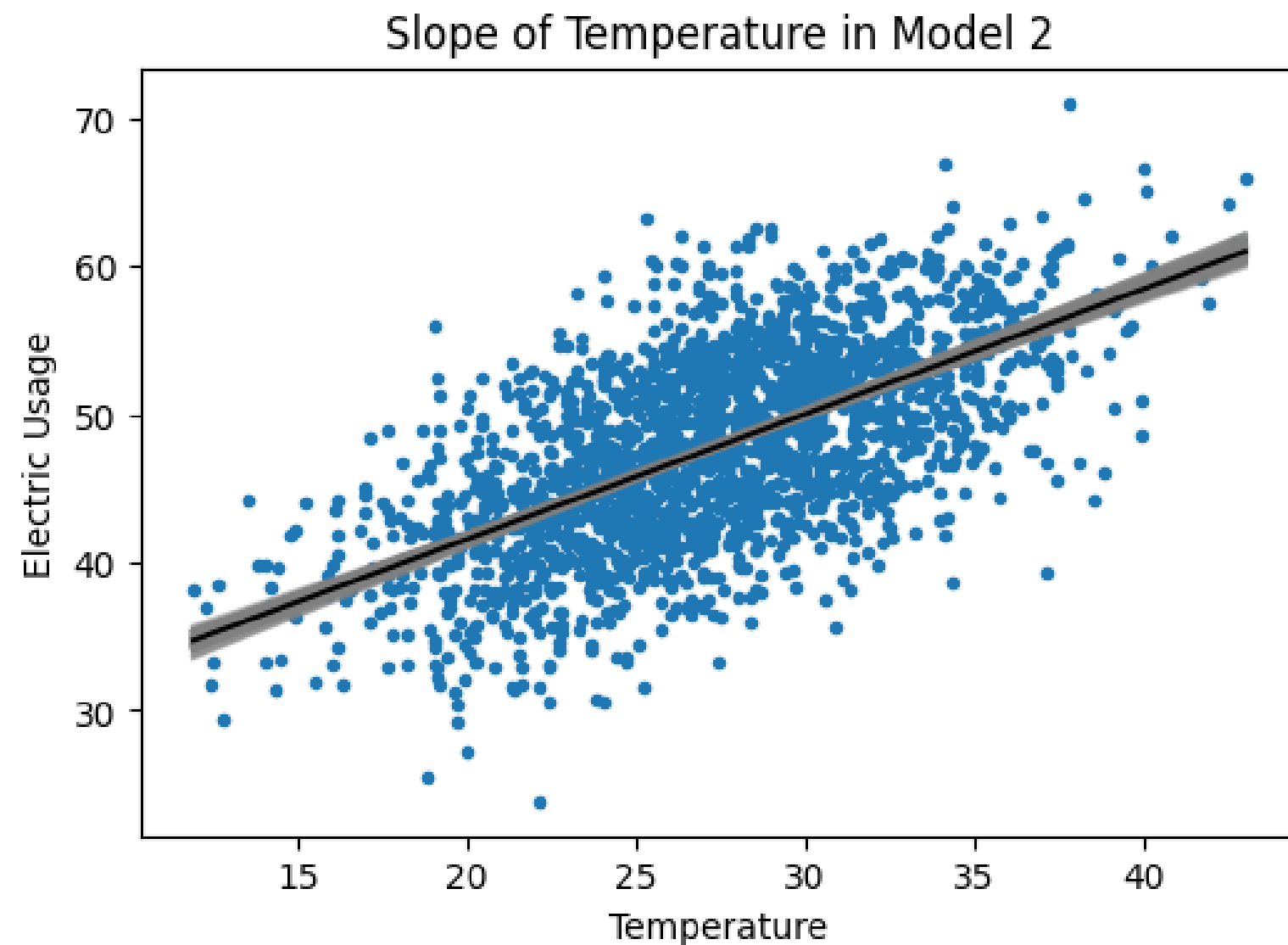
safe

REGRESSION MODELS - STANDARD ERROR COMPARISON

How collinearity inflates uncertainty in coefficient estimates

Model 1	Model 2	Model 3
Baseline	+ Solar Radiation	+ AC Usage
Electricity ~ Temperature	Electricity ~ Temp + Solar_Rad	Electricity ~ Temp + A_Usage
Std Err (Temperature) 0.0237	Std Err (Temperature) 0.0328	Std Err (Temperature) 0.0152
BASELINE	CAUTION	STABLE
Reference model. Only Temperature as predictor	Solar & Temperature correlated ($r = 0.69$). Std err increase	AC & Temperature independent ($r = 0.03$). Std err decrease

REGRESSION MODELS VISUALIZATION



From the visualization we see that the shadow around the line shows the result of regression line after repeatedly fit on the same data

EXAMPLE MULTICOLLINEARITY

Model 4

Extreme Multicollinearity

Electricity ~ Temp + Dew Point

$r = 0.995$

These two variables carry almost identical information. Including both destroys model stability.

What Extreme Collinearity Causes

Bootstrap Lines Scatter Wildly

- Repeated sampling produces hugely varying regression slopes — the model is unstable across samples.

Coefficient Signs May Flip

- One variable might show a positive and another negative coefficient, even when both should be positive.

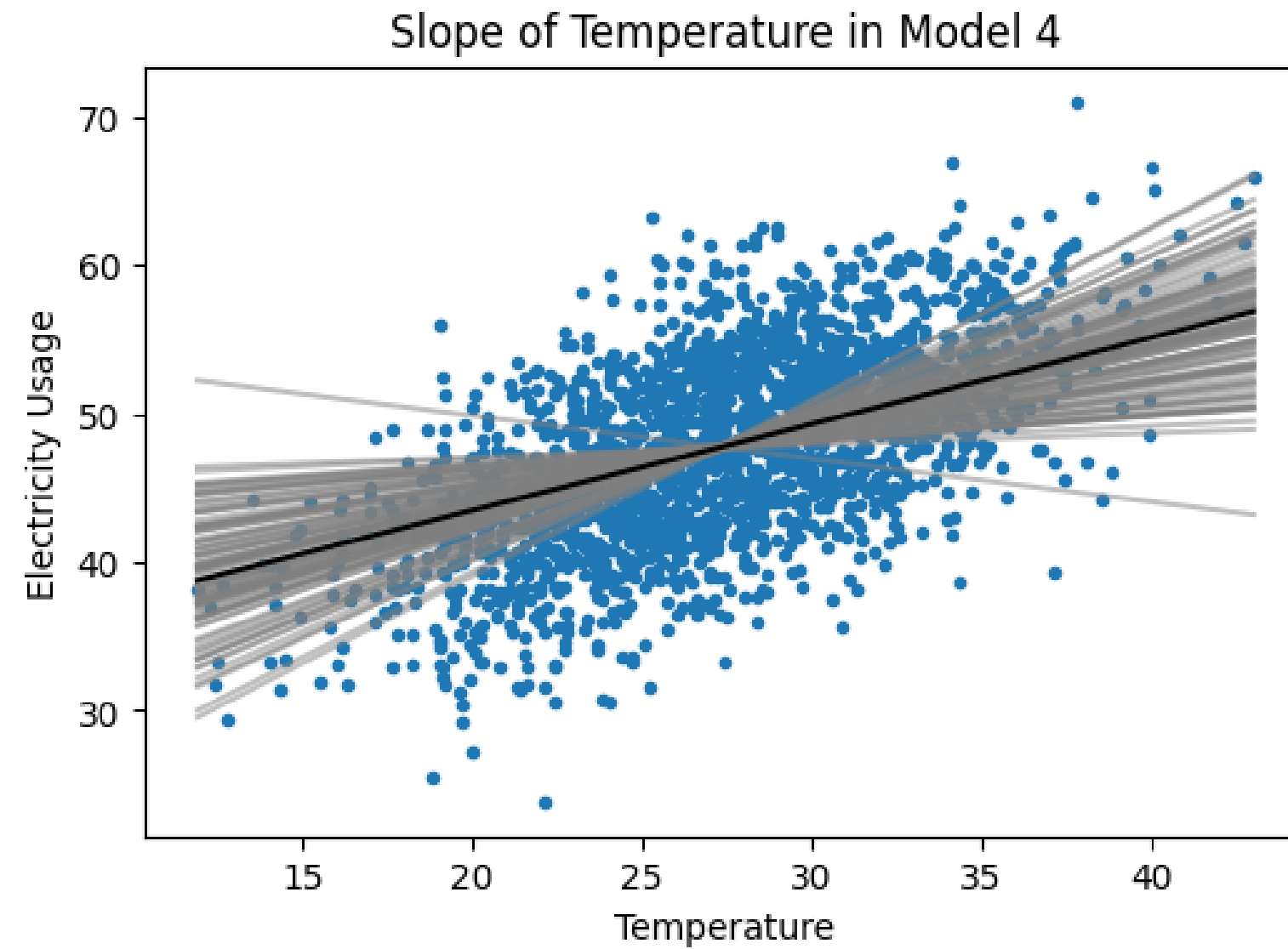
Standard Errors Explode

- Uncertainty in estimates becomes so large that no variable appears statistically significant.

Remove One Variable

- Solution: drop Dew Point (or Temperature). Keep the one that makes more theoretical sense.

REGRESSION MODELS VISUALIZATION



CONCLUSION

1

Collinearity between Temperature and Dew Point ($r = 0.995$) is extreme — include only one in any model.

2

Adding Solar Radiation ($r = 0.69$ with Temp) moderately inflates std error: $0.0237 \rightarrow 0.0328$.

3

Adding AC Usage ($r = 0.03$ with Temp) actually improves estimation stability: $0.0237 \rightarrow 0.0152$.

4

Always check collinearity with heatmaps and VIF before finalizing any regression model.



THANK YOU FOR YOUR TIME



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DATA ANALYST PRESENTATION
